Finite Sibling-Interface Descriptors and a Proposed Completion for $\mathcal{ALCI}_{RCC5}$

A proof-style note

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Abstract

This note continues the revised sibling-interface proposal for $\mathcal{ALCI}_{RCC5}$ and turns its open patchwork conjecture into an explicit theorem at the level of finite-prefix disjunctive networks. The key correction from the earlier note is retained: after splitting common descendants into EQ-mates, PP is no longer an open sibling cross-edge option. The post-split interface is organized into three statuses: Core (common descendants, handled by EQ), Out (already forced to be DR from the opposite subtree), and Front (still in the genuinely open DR/PO zone). We define a locally coherent finite descriptor language, assign disjunctive RCC5 domains to all finite partial unfoldings, prove non-emptiness under arc-consistency by a least-common-ancestor induction, and then use full tractability of disjunctive RCC5 networks plus K"onig's lemma to obtain a weak-EQ model. Quotienting by EQ yields a strong-EQ model. Combined with the finite-index lemma and the standard extraction of rank-d states, where $d = \mathrm{md}(C_0)$, this gives a finite quotient search procedure for concept satisfiability. The note is intended as a proof-style completion of the revised sibling-interface program; the extraction of witness menus from a model is condensed but finite throughout.

1 Context

Wessel's report leaves the decidability of $\mathcal{ALCI}_{RCC5}$ open and already isolates the real difficulty: models are complete RCC5-labelled graphs rather than ordinary trees, and infinite behaviour is driven by proper-part chains rather than plain modal branching [1]. More recent quotient-style approaches isolate the same obstruction from the opposite direction: exact extraction behaves cleanly for DR, PP, and PPI, while the remaining difficulty is the PO-sensitive cross-branch interface [2]. The revised sibling-interface note sharpened this picture by correcting an important mistake: after splitting join nodes into EQ-copies, PP is not an open sibling cross-edge value anymore. It is part of the common core and must be handled by splitting, not by cross-edge choice.

The present note takes the next step. It keeps the corrected three-way partition

$$\mathsf{Core}, \qquad \mathsf{Out}, \qquad \mathsf{Front},$$

and replaces the earlier open conjecture by an explicit finite-prefix patchwork theorem. The theorem is then combined with the finite-index lemma from the revised note to obtain a full quotient-to-model argument.

2 Preliminaries

2.1 RCC5 conventions

We use the non-EQ RCC5 relations

$$\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\} = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}.$$

The composition table is read in the standard way: if $x\,R\,y$ and $y\,S\,z$, then $x\,T\,z$ for some

$$T \in \mathrm{comp}(R, S).$$

We use only the following facts repeatedly:

$$\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \qquad \mathrm{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\},$$

$$\mathrm{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}, \qquad \mathrm{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\},$$

$$\mathrm{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\},$$

and for all $R, S \in \{\mathsf{DR}, \mathsf{PO}\}$,

$$\mathrm{comp}(R, S) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing.$$

We also use the full tractability of disjunctive RCC5 networks: if arc-consistency does not empty any domain, then a globally consistent atomic refinement exists [2, 1].

2.2 Types and modal depth

For an input concept C_0, let $\mathrm{cl}(C_0)$ be its Fischer–Ladner closure and let $\mathrm{Tp}(C_0)$ be the finite set of Hintikka types over $\mathrm{cl}(C_0)$. Write $d := \mathrm{md}(C_0)$ for the modal depth.

Definition 2.1 (Type-safe open labels)**.** *For Hintikka types $\tau, \upsilon \in \mathrm{Tp}(C_0)$, define*

$$\mathrm{Safe}_{\mathsf{DR}/\mathsf{PO}}(\tau, \upsilon) := \left\{ R \in \{\mathsf{DR}, \mathsf{PO}\} \;\middle|\; \begin{array}{l} \forall\,(\forall R.C \in \tau \Rightarrow C \in \upsilon), \\ \forall\,(\forall\,\mathrm{inv}(R).C \in \upsilon \Rightarrow C \in \tau) \end{array} \right\}.$$

If p, q are rank-k states for any k, write $\lambda(p)$ and $\lambda(q)$ for their local type components and set

$$\mathrm{Safe}_{\mathsf{DR}/\mathsf{PO}}(p, q) := \mathrm{Safe}_{\mathsf{DR}/\mathsf{PO}}(\lambda(p), \lambda(q)).$$

Definition 2.2 (Per-relation need families)**.** *For a Hintikka type $\tau \in \mathrm{Tp}(C_0)$ and a relation $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, define*

$$\mathrm{Need}_R(\tau) := \{\, C \in \mathrm{cl}(C_0) \mid \forall R.C \in \tau \,\}.$$

If p is a rank-k state for any k, write

$$\mathrm{Need}_R(p) := \mathrm{Need}_R(\lambda(p)).$$

A candidate RCC5 label $S \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ on an edge from a state p to a state q is called relation-safe *if*

$$\mathrm{Need}_S(p) \subseteq \lambda(q) \qquad \textit{and} \qquad \mathrm{Need}_{\mathrm{inv}(S)}(q) \subseteq \lambda(p).$$

For $S \in \{\mathsf{DR}, \mathsf{PO}\}$ this agrees with Definition 2.1. The point of the full family Need_R is that witness menus may introduce explicit DR/PO edges whose induced *pair-domains acquire candidate labels in $\{\mathsf{PP}, \mathsf{PPI}\}$ after composition with tree edges; those induced labels must also be filtered against the appropriate universal requirements.*

3 Split-tree presentations

Definition 3.1 (Weak-EQ split-tree presentation)**.** *A* weak-EQ split-tree presentation *of a candidate model of C_0 consists of:*

(i) a rooted tree T with ambient root ϱ_T and a distinguished evaluation node *$r_* \in T$,*

(ii) a surjective map $\mathrm{orig} : T \to D$ onto an underlying unsplit proper-part DAG D,

(iii) a local type assignment $\lambda : T \to \mathrm{Tp}(C_0)$,

such that:

- *tree edges are interpreted as immediate PPI edges downward and hence PP upward,*
- *if a node of D has several immediate predecessors, it may be represented by several copies in T,*
- *two tree nodes are in relation EQ exactly when they are distinct copies of the same original DAG node.*

All indirect PP/PPI edges are recovered from the ancestor relation in T.

Remark 3.2. This is a certificate presentation, not yet a genuine complete RCC5 model. The missing data are the labels on incomparable pairs of tree nodes. The whole point of the descriptor calculus is to make those labels finite and patchwork-solvable. The evaluation node r_* need not coincide with the ambient root ϱ_T; this is essential for formulas beginning with $\exists\mathsf{PP}.\top$, which are naturally evaluated at an internal tree node having an ancestor above it.

Definition 3.3 (Ordered sibling pair)**.** *Every node u of T is equipped with an arbitrary but fixed local total order $\lhd_u$ on its children. Two distinct children s, t of u form an* ordered sibling pair *(s, t). The order is not semantic; it only fixes the left/right orientation of the interface descriptor. Changing the local child order merely renames the left/right descriptor tables and does not change satisfiability. Both (s, t) and (t, s) are allowed as ordered sibling pairs, and when two endpoints x, y first diverge below u through children s, t, the ordered pair used in the network construction is the one determined by endpoint placement: $x \in T_s$ and $y \in T_t$. Write T_s and T_t for the rooted subtrees below them.*

4 The corrected status partition

From now on let (s, t) be an ordered sibling pair with

$$\rho(s, t) = \mathsf{PO}.$$

The DR case is rigid and handled separately below.

Definition 4.1 (Core, outside, frontier statuses)**.** *For $x \in T_s$ define:*

- $x \in \mathsf{Core}(s, t)$ *iff* $\mathrm{orig}(x)$ *lies below both* $\mathrm{orig}(s)$ *and* $\mathrm{orig}(t)$ *in the unsplit DAG;*
- $x \in \mathsf{Out}(s, t)$ *iff* $x \notin \mathsf{Core}(s, t)$ *and* $\rho(x, t) = \mathsf{DR}$*;*
- $x \in \mathsf{Front}(s, t)$ *iff* $x \notin \mathsf{Core}(s, t)$ *and* $\rho(x, t) = \mathsf{PO}$*.*

Define $\mathsf{Core}(t, s)$*,* $\mathsf{Out}(t, s)$*, and* $\mathsf{Front}(t, s)$ *symmetrically.*

Lemma 4.2 (Partition lemma)**.** *For every $x \in T_s$, exactly one of*

$$x \in \mathsf{Core}(s, t), \qquad x \in \mathsf{Out}(s, t), \qquad x \in \mathsf{Front}(s, t)$$

holds. Symmetrically for every $y \in T_t$.

Proof. If $x \in \mathsf{Core}(s, t)$ there is nothing to prove. So assume $x \notin \mathsf{Core}(s, t)$. Then x is distinct from t and not below it. Hence EQ is impossible, and PP is impossible because it would put x below t. The relation PPI is also impossible: it would put t below x below s, contradicting the incomparability of the sibling roots. RCC5 is jointly exhaustive on distinct nodes, so only DR or PO remain. Thus x belongs to exactly one of $\mathsf{Out}(s, t)$ or $\mathsf{Front}(s, t)$. The proof on the right side is symmetric. □

Remark 4.3. This is the crucial correction to the earlier note. After splitting, PP is no longer an open sibling cross-edge possibility; it is precisely the common-descendant core.

5 Rigid DR behaviour

Lemma 5.1 (Uniform DR below a DR root interface)**.** *If an ordered sibling pair (s, t) satisfies $\rho(s, t) = \mathsf{DR}$, then every pair of descendants $x \in T_s$, $y \in T_t$ satisfies*

$$\rho(x, y) = \mathsf{DR}.$$

Proof. Since $x \in T_s$, we have $x \,\mathsf{PP}\, s$. Since $s \,\mathsf{DR}\, t$, the singleton law $\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ gives $x \,\mathsf{DR}\, t$. Since $t \,\mathsf{PPI}\, y$, the singleton law $\mathrm{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ yields $x \,\mathsf{DR}\, y$. □

Proposition 5.2 (Outside status forces DR)**.** *Let (s, t) be an ordered sibling pair with $\rho(s, t) = \mathsf{PO}$.*

(a) If $x \in \mathsf{Out}(s, t)$, then for every $y \in T_t$ one has $\rho(x, y) = \mathsf{DR}$.

(b) If $y \in \mathsf{Out}(t, s)$, then for every $x \in T_s$ one has $\rho(x, y) = \mathsf{DR}$.

Proof. We prove (a). By definition, $x \,\mathsf{DR}\, t$. Since every $y \in T_t$ satisfies $t \,\mathsf{PPI}\, y$, the singleton law $\mathrm{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ yields $x \,\mathsf{DR}\, y$. The other direction is symmetric. □

Lemma 5.3 (Open cross-edge lemma)**.** *Let (s,t) satisfy $\rho(s,t) = \mathsf{PO}$. If*

$$x \in \mathsf{Out}(s,t) \cup \mathsf{Front}(s,t), \qquad y \in \mathsf{Out}(t,s) \cup \mathsf{Front}(t,s),$$

then

$$\rho(x,y) \in \{\mathsf{DR}, \mathsf{PO}\}.$$

Thus no open cross-edge between distinct non-core descendants can carry EQ*,* PP*, or* PPI*.*

Proof. If $\rho(x,y) = \mathsf{EQ}$, then $\mathrm{orig}(x) = \mathrm{orig}(y)$ and both nodes lie in the common core, contradiction. If $\rho(x,y) = \mathsf{PP}$, then x lies below y and therefore below t, so $x \in \mathsf{Core}(s,t)$, contradiction. If $\rho(x,y) = \mathsf{PPI}$, then symmetrically $y \in \mathsf{Core}(t,s)$, contradiction. RCC5 is jointly exhaustive on distinct pairs, so only DR and PO remain. □

6 Status evolution and open domains

Proposition 6.1 (Status automaton)**.** *Let (s,t) be an ordered sibling pair with $\rho(s,t) = \mathsf{PO}$, and let x' be a child of x in T_s. Then*

$$\mathsf{Core} \to \{\mathsf{Core}\}, \qquad \mathsf{Out} \to \{\mathsf{Out}\}, \qquad \mathsf{Front} \to \{\mathsf{Out}, \mathsf{Front}, \mathsf{Core}\}.$$

The same transition system holds symmetrically on the right.

Proof. If $x \in \mathsf{Core}(s,t)$, then every descendant of x remains below both sibling roots, so $x' \in \mathsf{Core}(s,t)$. If $x \in \mathsf{Out}(s,t)$, then $x \,\mathsf{DR}\, t$, and because $x' \,\mathsf{PP}\, x$, the singleton law $\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ yields $x' \,\mathsf{DR}\, t$, so $x' \in \mathsf{Out}(s,t)$. Finally, if $x \in \mathsf{Front}(s,t)$, then $x \,\mathsf{PO}\, t$ and $x' \,\mathsf{PP}\, x$, so

$$\rho(x',t) \in \mathrm{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}.$$

The values DR and PO place x' into $\mathsf{Out}(s,t)$ or $\mathsf{Front}(s,t)$, and the value PP places x' into the core. □

The open part of the descriptor language uses the three nonempty disjunctive domain values

$$\mathbb{D} := \{\{\mathsf{DR}\}, \{\mathsf{PO}\}, \{\mathsf{DR}, \mathsf{PO}\}\}.$$

The first value is rigid, the second is pure frontier overlap, and the third records unresolved DR/PO ambiguity.

Definition 6.2 (Rank-0 colors)**.** *Let*

$$\Sigma_0 := \mathrm{Tp}(C_0).$$

Definition 6.3 (Endpoint summaries)**.** *Fix an ordered sibling pair (s,t) and a rank $k \geq 0$. For $x \in T_s$, let $\mathrm{st}_k(x) \in \Sigma_k$ be its rank-k state (defined recursively below). Let $\mathrm{stat}_{s,t}(x) \in$*

$\{\mathsf{Core}, \mathsf{Out}, \mathsf{Front}\}$ *be the unique status of* x *relative to* (s,t) *from Lemma 4.2. The* left endpoint summary *of* x *with respect to* (s,t) *is the pair*

$$\chi_{s,t}(x) := (\mathrm{st}_k(x), \mathrm{stat}_{s,t}(x)).$$

Thus the first component $p := \mathrm{st}_k(x)$ *is a rank-k state, and the second component* $\sigma := \mathrm{stat}_{s,t}(x)$ *is the endpoint status. If* $\sigma = \mathsf{Core}$*, then* $K_k(s,t)(p) \neq 0$*; if* $\sigma \in \{\mathsf{Out}, \mathsf{Front}\}$*, then* $(p,\sigma) \in L_k(s,t)$*. The* right endpoint summary $\chi_{t,s}(y)$ *for* $y \in T_t$ *is defined symmetrically.*

Definition 6.4 (Rank-k descriptor)**.** *Assume* Σ_k *finite. For an ordered sibling pair* (s,t) *with* $\rho(s,t) = \mathsf{PO}$*, its rank-k descriptor consists of:*

(i) a core multiplicity map

$$K_k(s,t) : \Sigma_k \to \{0, 1, +\};$$

(ii) left and right open supports

$$L_k(s,t), R_k(s,t) \subseteq \Sigma_k \times \{\mathsf{Out}, \mathsf{Front}\};$$

(iii) an open relation table

$$M_k(s,t) : L_k(s,t) \times R_k(s,t) \to \mathbb{D};$$

(iv) bounded multiplicity maps on L_k *and* R_k *with values in* $\{0, 1, +\}$*.*

For a DR*-root interface, we use the trivial descriptor forced by Lemma 5.1.*

Definition 6.5 (Local coherence axioms)**.** *A rank-k descriptor is* locally coherent *if the following hold.*

(C1) If either endpoint status is Out*, then the table value is* $\{\mathsf{DR}\}$*.*

(C2) Every frontier-frontier entry indexed by endpoint summaries (p, Front) *and* (q, Front) *is a nonempty subset of*

$$\mathrm{Safe}_{\mathsf{DR}/\mathsf{PO}}(p,q).$$

In other words, every open label carried by that table entry is type-safe for the local types of the two endpoint states.

(C3) (Left downward support) If a left endpoint (p,σ) *can evolve in one child step to* (p',σ')*, then for every right endpoint* (q,τ) *and every*

$$r \in M_k((p,\sigma),(q,\tau))$$

there exists

$$r' \in M_k((p',\sigma'),(q,\tau))$$

with

$$r' \in \mathrm{comp}(\mathsf{PP}, r),$$

and conversely for every such r' there exists r with

$$r \in \mathrm{comp}(\mathsf{PPI}, r').$$

(C4) (Right downward support) The symmetric condition holds when the right endpoint evolves by one child step.

(C5) (Triple coherence) For every parent node, every three child interfaces emanating from that parent, and every compatible choice of endpoint colors/statuses from the three pairwise tables, each pair of values in the first two tables has support in the third table via the RCC5 composition table.

Remark 6.6. The role of (C3) and (C4) is exactly to forbid the unsupported pattern

$$M = \{\mathsf{DR}, \mathsf{PO}\} \quad \text{above}, \qquad M' = \{\mathsf{PO}\} \quad \text{below}$$

when the DR value has no PP-support on the lower edge. This is the point where the crude maximal-domain construction fails and where the descriptor table must already carry support information.

7 Ranked node states and finite index

Definition 7.1 (Rank-$(k+1)$ state)**.** *Assume rank-k colors are given. The rank-$(k+1)$ state of a node u is*

$$\mathrm{State}_{k+1}(u) := (\lambda(u), \mathrm{Par}_k(u), \mathrm{Child}_k(u), \mathrm{Iface}_k(u), \mathrm{Wit}_k(u)),$$

where:

- $\lambda(u) \in \mathrm{Tp}(C_0)$ *is the local type,*
- $\mathrm{Par}_k(u) \in \Sigma_k \cup \{\bot\}$ *records the unique parent color of u when $u \neq \varrho_T$, and $\bot$ at the ambient tree root,*
- $\mathrm{Child}_k(u) : \Sigma_k \to \{0, 1, +\}$ *records the multiplicities of rank-k child colors, where $+ =$ "at least two" (not "infinitely many"),*
- $\mathrm{Iface}_k(u)$ *is the finite family of locally coherent rank-k sibling descriptors between ordered child pairs of u,*
- $\mathrm{Wit}_k(u)$ *is a finite witness menu for the existential formulas in $\lambda(u)$, recording for each existential one designated target color and target relation domain together with the finite relation-sensitive side constraints saying that every explicit or induced candidate label $S \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ appearing in that menu is relation-safe in the sense of Definition 2.2.*

The coarse multiplicity alphabet $\{0, 1, +\}$ is used only to distinguish absence, uniqueness, and non-uniqueness of child colors; the infinitary behaviour of the final model comes from unbounded depth of the unfolding, not from infinite branching at a node. Let Σ_{k+1} be the set of all such states.

Definition 7.2 (Rank-d state)**.** *An element of Σ_d is called a* rank-d state*. More generally, if x is a node occurring in a model, split-tree presentation, or unfolding, then $\mathrm{st}_d(x) \in \Sigma_d$ denotes the rank-d state attached to x.*

Theorem 7.3 (Finite-index lemma)**.** *For every $k \geq 0$, the set Σ_k is finite.*

Proof. The proof is by induction. The base case $\Sigma_0 = \mathrm{Tp}(C_0)$ is finite because $\mathrm{cl}(C_0)$ is finite. For the inductive step, assume Σ_k finite. Then there are only finitely many parent slots Par_k, finitely many child multiplicity maps Child_k, finitely many support sets L_k, R_k, finitely many core profiles K_k, finitely many table values in $\mathbb{D}$, finitely many locally coherent relation tables M_k, and finitely many bounded multiplicity maps on supports. The witness menu ranges over the finite closure and the finite alphabet of rank-k colors and open domain values. Hence only finitely many tuples

$$(\lambda, \mathrm{Par}_k, \mathrm{Child}_k, \mathrm{Iface}_k, \mathrm{Wit}_k)$$

exist. Therefore Σ_{k+1} is finite. □

8 Finite quotients and finite prefixes

Definition 8.1 (Valid finite rank-d quotient)**.** *A* valid finite rank-d quotient *for C_0 is a tuple*

$$Q = (S, \sigma^{\uparrow}, \sigma_*, \mathrm{Par}_d, \mathrm{Child}_d, \mathrm{Iface}_d, \mathrm{Wit}_d),$$

where:

- *$S \subseteq \Sigma_d$ is a finite set of rank-d states,*
- *$\sigma^{\uparrow} \in S$ is a distinguished* ambient root state*,*
- *$\sigma_* \in S$ is a distinguished* evaluation state*, not necessarily equal to $\sigma^{\uparrow}$,*
- *the parent slot, child multiplicity data, sibling-interface descriptors, and witness menus are those inherited from the states in S, using only targets that again lie in S.*

These data must satisfy: every existential requirement in every state's local type is matched by the witness menu; every universal requirement is respected by the type-safe relation domains in the frontier descriptors; and, crucially, every explicit witness-menu edge and every induced pair-domain created from it by composition with tree edges is filtered relation-sensitively. Concretely, whenever a candidate label $S \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ remains available between endpoint states p and q, one must have

$$\mathrm{Need}_S(p) \subseteq \lambda(q) \qquad \textit{and} \qquad \mathrm{Need}_{\mathrm{inv}(S)}(q) \subseteq \lambda(p).$$

Repeated state occurrences are quotiented by equality of rank-d states.

Remark 8.2 (Why the full need family matters)*.* Although the only genuinely open frontier-frontier labels are DR and PO, witness menus can still generate induced pair-domains whose candidate labels include PP or PPI after composition with tree edges. A simple example is

$$(\forall \mathsf{PO}.\neg C) \sqcap (\forall \mathsf{DR}.\neg C) \sqcap (\forall \mathsf{PP}.\neg C) \sqcap (\forall \mathsf{PPI}.\neg C) \sqcap \exists \mathsf{DR}(\exists \mathsf{PO}.C),$$

which is unsatisfiable: if $x \xrightarrow{\mathsf{DR}} y \xrightarrow{\mathsf{PO}} z$ with $z \models C$, then the induced domain on (x, z) is obtained from $\mathrm{comp}(\mathsf{DR}, \mathsf{PO})$ and every candidate label in that domain forces $z \models \neg C$. A quotient or tableau side-checker that only remembers $\forall\mathsf{DR}$- and $\forall\mathsf{PO}$-requirements would miss this contradiction. The relation-sensitive Need_R family is therefore part of the semantic validity condition.

Remark 8.3. For the present note, the quotient is taken as the finite certificate produced by the state language. The model-guided extraction of such quotients is standard in the global-caching tradition: one reads off rank-d states from a satisfying split-tree presentation and identifies equal states. The only genuinely new soundness issue is the complete-graph cross-edge assignment; that is what is proved next.

Definition 8.4 (Unfolding $U(Q)$)**.** *Let Q be a valid finite rank-d quotient. Its unfolding $U(Q)$ is the pointed split tree obtained as follows.*

(i) Start with a fresh copy of the evaluation state σ_*; this copy is the distinguished evaluation node of $U(Q)$.

(ii) Realize child multiplicity requirements and witness-menu entries downward by fresh copies of the corresponding target states.

(iii) Realize the unique predecessor slot upward until a node with parent slot $\perp$ is reached; that node is the ambient root copy of $U(Q)$ and has state $\sigma^{\uparrow}$.

(iv) Iterate the downward construction at every generated node.

Core matches are unfolded as distinct EQ*-mates. These mates are different tree copies of one intended semantic point. They are generated from the same rank-d state and inherit the same witness menu; witness obligations are interpreted at the level of the intended* EQ*-class rather than at one designated copy. A* finite prefix *of $U(Q)$ is a finite connected pointed subtree containing the distinguished evaluation node and closed under predecessors.*

Definition 8.5 (Disjunctive network of a finite prefix)**.** *Let P be a finite prefix of $U(Q)$. We define a disjunctive RCC5 network $N(P)$ on the nodes of P.*

(i) If $x = y$, set $D(x, y) = \{\mathsf{EQ}\}$.

(ii) If x and y are EQ*-mates, set $D(x, y) = \{\mathsf{EQ}\}$.*

(iii) If x is an ancestor of y, set $D(x, y) = \{\mathsf{PPI}\}$ and $D(y, x) = \{\mathsf{PP}\}$.

(iv) If x and y first diverge at an ordered sibling pair (s, t), use the descriptor of (s, t) as follows. Let

$$\chi_{s,t}(x) = (p, \sigma), \qquad \chi_{t,s}(y) = (q, \tau)$$

be their endpoint summaries from Definition 6.3. Thus p and q are rank-k states, while σ and τ are the unique endpoint statuses. If $\sigma = \mathsf{Core}$, then $K_k(s, t)(p) \neq 0$; if $\sigma \in \{\mathsf{Out}, \mathsf{Front}\}$, then $(p, \sigma) \in L_k(s, t)$. The right-hand side is analogous. Then:

- *if either endpoint is in the core, replace that endpoint by its designated mate in the opposite branch and recurse;*

- *if either endpoint status is* Out*, set* $D(x, y) = \{\mathsf{DR}\}$*;*
- *if both endpoint statuses are* Front*, set*

$$D(x, y) = M_k(s, t)((p, \mathsf{Front}), (q, \mathsf{Front})).$$

The converse domain $D(y, x)$ *is determined by inversion. In particular, if* x *and* y *are* EQ*-mates, then the designated-mate recursion guarantees that for every third node* z *one has*

$$D(x, z) = D(y, z), \qquad D(z, x) = D(z, y).$$

Thus EQ*-mates are synchronized already at the level of disjunctive domains.*

9 Finite-prefix arc-consistency theorem

Theorem 9.1 (Finite-prefix non-emptiness)**.** *Let* Q *be a valid finite rank-d quotient. For every finite prefix* P *of* $U(Q)$*, arc-consistency applied to the disjunctive network* $N(P)$ *of Definition 8.5 never empties a domain.*

Proof. We prove the stronger statement that every domain value surviving the local construction has support through every third node in P. The proof is by induction on the maximum height of the least common ancestors appearing in triples of nodes.

Case 1: one pair is an EQ pair. If $D(x, y) = \{\mathsf{EQ}\}$, then by construction x and y have identical local type and identical domains to every third node z. Since

$$\mathrm{comp}(\mathsf{EQ}, R) = \{R\} = \mathrm{comp}(R, \mathsf{EQ}),$$

every support check is immediate.

Case 2: all three nodes lie in one child subtree. Then the whole triple belongs to a strictly smaller finite prefix, so the induction hypothesis applies.

Case 3: the three nodes lie in three distinct child subtrees of one parent. After reducing core endpoints to their designated mates, every pair-domain is one of

$$\{\mathsf{DR}\}, \qquad \{\mathsf{PO}\}, \qquad \{\mathsf{DR}, \mathsf{PO}\}.$$

By local triple coherence (C5), every pair of values chosen from the first two domains has support in the third. Hence no value is removed.

Case 4: two nodes lie in one child subtree and the third lies in a sibling subtree. Assume $x, z \in T_s$ and $y \in T_t$ for an ordered sibling pair (s, t).

Subcase 4a: x and z are incomparable inside T_s. Then their domain is already controlled by a strictly lower sibling descriptor inside T_s, so the induction hypothesis reduces the problem to Case 3.

Subcase 4b: x is above z. Then

$$D(x, z) = \{\mathsf{PPI}\}, \qquad D(z, x) = \{\mathsf{PP}\}.$$

Write

$$X := D(x, y), \qquad Y := D(z, y).$$

By the status automaton and the support axioms (C3)–(C4), the pair (X, Y) is one of the allowed downward-support configurations. We must check the two directed support conditions.

For the triangle $x \to z \to y$, every value in Y has an ancestor support in X by the second half of (C3). Equivalently, every value in X has a child support in Y through the composition with PP required by the first half of (C3). This is exactly the support needed for arc-consistency on the triple (z, x, y).

For the triangle $z \to x \to y$, the symmetric support through PPI is guaranteed by the converse half of (C3). Concretely, the only composition facts used are

$$\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}, \qquad \mathrm{comp}(\mathsf{PP}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\},$$

$$\mathrm{comp}(\mathsf{PPI}, \mathsf{DR}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PPI}\}, \qquad \mathrm{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}.$$

The local support axiom was chosen exactly so that unsupported combinations such as

$$X = \{\mathsf{DR}, \mathsf{PO}\}, \qquad Y = \{\mathsf{PO}\}$$

are forbidden whenever the DR value would have no PP-support below. Hence no value is removed in this subcase either.

The case where z is above x is symmetric.

Since every triple falls under one of the four cases, every domain retains support through every third node. Hence arc-consistency empties no domain. □

Remark 9.2. The theorem is intentionally stated in the arc-consistency form needed by full tractability, not as a claim that the raw domains are automatically path-complete before pruning. The descriptor language carries exactly the additional support information needed to prevent empty domains under pruning.

10 Prefix extension and canonical refinement

Fix once and for all an exhaustion

$$P_0 \subseteq P_1 \subseteq P_2 \subseteq \cdots \subseteq U(Q)$$

of the unfolding by finite prefixes such that each P_n is closed under predecessors and under EQ-mates.

Definition 10.1 (Extendible atomic refinement)**.** *Let P_n be a finite prefix. An atomic refinement A_n of $N(P_n)$ is called* extendible *if for every $m \geq n$ there exists an atomic refinement A_m of $N(P_m)$ such that*

$$A_m \restriction_{P_n} = A_n.$$

Thus an extendible refinement is one whose current DR/PO *choices can be continued along the fixed exhaustion without later backtracking.*

Lemma 10.2 (Existence of extendible refinements)**.** *For every n, the finite prefix network $N(P_n)$ has at least one extendible atomic refinement.*

Proof. Consider the tree whose nodes at level $m \geq n$ are the atomic refinements of $N(P_m)$, ordered by restriction. By Theorem 9.1 and full tractability of disjunctive RCC5 networks, every level is nonempty. Because every P_m is finite and every open edge-domain is one of $\{\mathsf{DR}\}$, $\{\mathsf{PO}\}$, or $\{\mathsf{DR}, \mathsf{PO}\}$, each level is finite; hence the tree is finitely branching. By K"onig's lemma there is an infinite branch through this tree. Its level-n node is an atomic refinement of $N(P_n)$ that extends to every later prefix, hence is extendible. □

Lemma 10.3 (Prefix-extension lemma)**.** *Let A_n be an extendible atomic refinement of $N(P_n)$. Then there exists an extendible atomic refinement A_{n+1} of $N(P_{n+1})$ such that*

$$A_{n+1} \restriction_{P_n} = A_n.$$

Proof. Consider the subtree of the refinement tree from Lemma 10.2 consisting only of atomic refinements of $N(P_m)$, $m \geq n$, that extend A_n. This subtree is finitely branching and has a node on every level because A_n is extendible. By K"onig's lemma it has an infinite branch. The level-$(n+1)$ node on that branch is an extendible atomic refinement of $N(P_{n+1})$ extending A_n. □

To make the construction deterministic, fix once and for all:

(i) a global enumeration of unordered node pairs in $U(Q)$ that is compatible with the exhaustion $P_0 \subseteq P_1 \subseteq \cdots$,

(ii) and an order on the genuinely open labels,

$$\mathsf{DR} \prec \mathsf{PO}.$$

These choices induce, for every n, a lexicographic order $\prec_n$ on the atomic refinements of $N(P_n)$: compare two refinements on the first enumerated pair where they differ, and prefer DR to PO.

Lemma 10.4 (Canonical refinements)**.** *For every n, there exists a unique $\prec_n$-least extendible atomic refinement*

$$\mathrm{Can}_n$$

of $N(P_n)$. Moreover,

$$\mathrm{Can}_{n+1} \restriction_{P_n} = \mathrm{Can}_n$$

for all n.

Proof. By Lemma 10.2, the finite set of atomic refinements of $N(P_n)$ contains at least one extendible element. Hence a $\prec_n$-least extendible element exists, and uniqueness is immediate from linearity of the lexicographic order.

For compatibility, let

$$B_n := \mathrm{Can}_{n+1} \restriction_{P_n} .$$

Since Can_{n+1} is extendible, so is B_n. If $B_n \neq \mathrm{Can}_n$, then by minimality of Can_n one has

$$\mathrm{Can}_n \prec_n B_n.$$

By Lemma 10.3, Can_n has an extendible refinement C_{n+1} to P_{n+1}. By construction of the lexicographic order, the first pair on which C_{n+1} and Can_{n+1} can differ must already witness

$$C_{n+1} \prec_{n+1} \mathrm{Can}_{n+1},$$

contradicting the minimality of Can_{n+1}. Therefore $B_n = \mathrm{Can}_n$. □

Definition 10.5 (Typed EQ-congruence)**.** *Let M be a weak-EQ RCC5 structure whose nodes carry the local types inherited from states. We say that* EQ *is a* typed congruence *on M if for all nodes x, y, z:*

(i) $x \,\mathsf{EQ}\, y$ implies that x and y have the same local type,

(ii) $x \,\mathsf{EQ}\, y$ implies $\rho(x, z) = \rho(y, z)$,

(iii) $x \,\mathsf{EQ}\, y$ implies $\rho(z, x) = \rho(z, y)$.

If these conditions hold, the quotient M/EQ is well defined as an RCC5 structure because every EQ*-class has a unique type and unique incoming/outgoing relation to every other class.*

Lemma 10.6 (EQ-synchrony in atomic refinements)**.** *Let P be a finite prefix of $U(Q)$ and let A be any atomic refinement of $N(P)$. If x and y are* EQ*-mates in P, then:*

(a) x and y have the same local type,

(b) for every $z \in P$, one has $A(x, z) = A(y, z)$ and $A(z, x) = A(z, y)$.

Proof. Part (a) holds because EQ-mates are created from the same rank-d state in Definition 8.4, hence inherit the same local type. For part (b), the network construction gives

$$D(x, y) = \{\mathsf{EQ}\}, \qquad D(x, z) = D(y, z), \qquad D(z, x) = D(z, y)$$

for every third node z. Since A is an atomic refinement, write

$$A(x, y) = \mathsf{EQ}, \qquad A(y, z) = R.$$

The triangle (x, y, z) must be composition-consistent, so

$$A(x, z) \in \mathrm{comp}(\mathsf{EQ}, R) = \{R\}.$$

Hence $A(x, z) = R = A(y, z)$. The converse direction is identical, using

$$A(z, y) = S \quad \Rightarrow \quad A(z, x) \in \mathrm{comp}(S, \mathsf{EQ}) = \{S\}.$$

Thus $A(z, x) = A(z, y)$ as required. □

11 From finite prefixes to a model

Theorem 11.1 (Weak-EQ realization)**.** *Every valid finite rank-d quotient Q unfolds to a weak-*EQ *model of C_0.*

Proof. Fix the exhaustion $P_0 \subseteq P_1 \subseteq P_2 \subseteq \cdots$ from the previous section and let

$$\mathrm{Can}_0 \subseteq \mathrm{Can}_1 \subseteq \mathrm{Can}_2 \subseteq \cdots$$

be the compatible sequence of canonical extendible refinements from Lemma 10.4. Define

$$\mathrm{Can}_\infty := \bigcup_{n<\omega} \mathrm{Can}_n \,.$$

Because the canonical refinements are compatible under restriction, Can_∞ labels every unordered pair of nodes in $U(Q)$ by a unique RCC5 base relation.

Every finite triple of nodes appears already in some finite prefix P_n, and Can_n is an atomic refinement of $N(P_n)$. Therefore every finite triple is composition-consistent under Can_∞, so the union is globally composition-consistent. Every designated existential requirement is satisfied because the unfolding was built from the witness menus. Every universal requirement is respected for two separate reasons: frontier-frontier table entries were constrained to be type-safe in Definition 6.5, and all explicit or induced witness-menu domains were required in Definition 8.1 to satisfy the full relation-sensitive conditions from Definition 2.2. Hence Can_∞ turns the unfolding into a weak-EQ model.

Moreover, Lemma 10.6 applies on every finite prefix and therefore to every finite restriction of Can_∞. It follows that the resulting EQ relation is a typed congruence in the sense of Definition 10.5. The distinguished evaluation node realizes C_0 by the usual induction on modal depth. □

Theorem 11.2 (Strong-EQ realization)**.** *Every valid finite rank-d quotient Q yields a strong-*EQ *model of C_0.*

Proof. By Theorem 11.1, Q has a weak-EQ realization M whose EQ relation is a typed congruence by Definition 10.5. Quotient M by the equivalence relation EQ. Because each EQ-class has a unique local type and unique incoming/outgoing RCC5 relation to every other class, the quotient structure M/EQ is well defined as an RCC5 model, now with strong identity semantics for EQ. Satisfaction of formulas in $\mathrm{cl}(C_0)$ is preserved under the quotient: existentials survive because a class has a witness as soon as one representative has one, while universals depend only on the type and the induced class-to-class edge labels. Therefore the distinguished evaluation class satisfies C_0. □

12 Decidability consequence

Theorem 12.1 (Decidability via finite quotients)**.** *Concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable.*

Proof sketch. Fix $d = \mathrm{md}(C_0)$. By Theorem 7.3, only finitely many rank-d states exist. Hence there are only finitely many finite quotient candidates up to isomorphism. Enumerate them and test the local validity conditions from Definitions 6.5 and 8.5 together with the witness-menu constraints.

If a valid quotient is found whose distinguished evaluation state's local type contains C_0, Theorems 11.1 and 11.2 produce a genuine model of C_0.

Conversely, from any pointed strong-EQ model (M, m_*) of C_0, construct a split-tree presentation by splitting join nodes into EQ-mates and reading off rank-d states. Choose an ambient root above m_* in the split tree; because $d = \mathrm{md}(C_0)$ is finite, only finitely many ancestor levels above the distinguished evaluation point are relevant to formulas in $\mathrm{cl}(C_0)$. Because only finitely many such rank-d states exist, equal states can be quotiented. The descriptor tables are extracted as the least locally coherent support-closed tables containing the realized atomic relations of the model. This yields a valid finite quotient. Therefore C_0 is satisfiable iff a valid finite quotient exists, and the search terminates. □

Remark 12.2. The only compressed part in the final theorem is the extraction of the support-closed tables and relation-sensitive witness-menu filters from a model. That extraction is finite because the closure, the rank-d color alphabet, and the open domain alphabet are all finite, and the Need_R families are read off directly from local types. The soundness direction proved above is the genuinely new global step.

13 Conclusion

The revised sibling-interface program becomes significantly sharper once PP is removed from the open sibling interface and absorbed into the split core. After that correction, the cross-edge problem decomposes cleanly:

- EQ is fixed by split copies,
- DR is rigid on all Out zones,
- the only unresolved open labels are DR/PO on frontier-frontier pairs.

The descriptor language records exactly the support information needed for those frontier pairs. The finite-prefix arc-consistency theorem then supplies the missing patchwork step, and full tractability of disjunctive RCC5 networks finishes the model-construction argument. In that sense, the present note is a proposed completion of the corrected split-forest proof strategy for $\mathcal{ALCI}_{RCC5}$.

References

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